

Session 6: Network Effects, Platforms and Standards

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Network externalities

Demand-side scale economies

Themes:

- Compatibility is valuable, standards emerge.
 - Competition becomes between standards.
- Multiple equilibria: inertia, positive feedback.
- New trade-offs: compatibility, openness.
 - Relationships w/other firms change.
 - Competitors and *complementors*.
- Platforms emerge as meeting points

Outline

1. Network effect: fundamentals
2. Standards
3. Multi-sided Platforms

1. Network Effects: Fundamentals

Network externalities exists when user i's willingness to pay for a product depends on how many others have adopted the same or compatible products.

When do these arise? Three types

- Direct effects.
- Indirect (Complements).
- Liquidity effects (market thickness)

Direct Effects

“Direct” network externalities are common in communication protocols.

- Telephones, fax machines.
- Instant messaging, languages.

Here user i cares directly whether user j adopts.

Complements Effects

User i might not care about others' adoption per se, but does care through its effect on the supply of complements.

Supply of movies on a particular format is influenced by popularity of the format.

- Software, music, etc.

Liquidity Effects

Two distinct type of users

Network effects go from one group to the other:

- Buyers want to participate in markets with many sellers, sellers want to participate in markets with many buyers.

- NYSE
- Ebay
- Yellow Pages
- Newspapers (via advertising)

Network Effects:

A basic coordination game

- New technology = social media

		<i>User 2</i>	
		Join	Don't
<i>User 1</i>	Join	(2,2)	(-1,0)
	Don't	(0,-1)	(0,0)

(To make this more concrete...)

Network effects: Game 1

- New technology = social media (Facebook)

		User 2	
		 Don't	Don't
User 1		(2,2)	(-1,0)
	Don't	(0,-1)	(0,0)

- *What would everyone prefer to be at?*
- *What is (are) the Nash equilibrium (equilibria)?*

Network effects: Game 1

- New technology = social media (Facebook)

		User 2	
		 Like	Don't
User 1	 Like	(2,2)	(-1,0)
	Don't	(0,-1)	(0,0)

- *Takeaway: need not get to adoption—even if its preferred*
- *Keep this in mind—it will affect how we manage the tech.*

Rival platforms: Game 2

- New technology = social media (Mastodon vs. Twitter)
- Suppose users can adopt only one

		User 2	
User 1			
	 	(2,2)	(-1,-1)
		(-1,-1)	(1,1)

- *What would everyone prefer to be at?*
- *What is/are Nash?*

Rival platforms: Game 2

- New technology = social media (Mastodon vs. Twitter)
- Suppose users can adopt only one

		User 2	
User 1		(2,2)	(-1,-1)
		(-1,-1)	(1,1)

- *Takeaway: need not adopt the preferred or superior technology*
- *Keep this in mind—we'll see this happen in the real world*

Lock in effects

-
- Markets with network effects can be stuck on inferior solutions because of disagreements on where to go next.
- Example: QWERTY
 - QWERTY keyboard designed to avoid jams in old technology (1873) by minimizing clashes between typebars
 - Dvorak keyboard, invented in 1932, faster.
 - But QWERTY standard reinforced by touch typing knowledge acquired by users

Adding dynamics on user side: Game 3

- Setup is the same as before

Player 1 adopts and player 2 doesn't	$(-1,0)$
Player 2 adopts and player 1 doesn't	$(0,-1)$
Neither player adopts	$(0,0)$
Both players adopt	$(2,2)$

- Suppose one user is an “**early adopter**” (i.e., they choose first)
- What is likely to be played?

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- Suppose one user is an “**early adopter**” (i.e., they choose first)
- What is likely to be played?
- *Takeaway: creating/harnessing early adopters ensures “good” eq.*
- *May explain part of the rise in bloggers, vloggers, and influencers*

Tipping

Users tend to follow each others' decisions.

- Markets can tip one way or the other.
- Changes are rare, but happen quickly when they do.

Positive feedback effects.

- Fax machines: basic technology A. Bain (1843); ATT first service 1925; explosive growth 1982-1987
- E-mail: available 1969; explosive growth from 1989 (double every year 89-95, then even faster growth)

Chicken-and-egg problem

- Simple game framework makes clear that
 - “Good” technology may not be adopted
 - “Best” technology may not be adopted
 - One user’s adoption choice depends on what he or she expects others to do (i.e., expectations play a big role!)
 - Firm success depends on **coordination**
- Strategies include
 - Creating and harnessing **early adopters**—perhaps at almost any cost
 - **Subsidizing** early periods (e.g., trial periods, direct “bribes”)

Managerial solutions

What does this mean for managers?

- Coordination is crucial
 - Often better to be chosen than to be higher quality
 - Often coordination effort more important than R&D
- Specifically...
 - Aggressively promote brand, product, use, interactions
 - Create and convince early adopters *at nearly any cost*
 - Subsidize in early periods and raise price later (i.e., trial periods and sign-up bonuses)

Expectations are key

Consumers expectations are key in adoption of new technology

- Apple/Android/Microsoft in Mobile OS
- Microsoft expert in ‘vaporware’ in the 1990s
- Best way to manage expectations is assembling allies around new technology

Options

- Segment key users, subsidize them directly
 - Adobe PDF
- Develop killer app targeted at them
 - Adobe PS
- Influence their expectations.
 - “aura of inevitability”

Dealing with Tipping

Leaders: facilitate this.

- Be ready to scale up quickly.

Laggards: impede this.

- Create confusion, negate “aura of inevitability.”
- Delays via other forums?
- Last ditch: preserve base, hope for tomorrow.

II. Standards

Standards

Standards are valuable when network externalities exist.

Standards may be market-determined...

- “*De facto* standards”
...or determined via regulation.
- “*De jure* standards”

Standards

The most important form of competition may be across standards (or platforms) rather than across firms.

- That is, the pressing industry structure question might be “how many standards” rather than “how many firms.”

How Many Standards?

Depends on how much users value compatibility relative to a standard's particular attributes.

- Time: one.
- Music playback: several.
- Automobiles: many.

Two key factors

- 1) How strong are network effects?
- 2) How strong is taste for variety?

How many standards?

Two Fundamental Forces:

1. Strength of Network Effects

- Do users need a large network for the product to work?
- Does value keep increasing with network size?

2. Taste for Variety

- How different are user preferences?
- How important are these differences?

Note: When we examine platforms in detail, we'll add two more factors that specifically affect platform competition

How likely is that a Dominant Standard will emerge?

	Weak Network Externalities	Strong Network Externalities
Low Demand For Variety	Unlikely	High
High Demand For Variety	Low	Depends

Compatibility v. Performance

Firms face a technological trade-off

Consumers prefer more of both

- $MRS = MRT$...completely standard to here.

But in markets with network externalities,

- Getting this right today provides you a larger installed base, and *a competitive advantage tomorrow*.
- Providing compatibility not easy when you do not control the interface.

Standards: Open v. Proprietary?

Value capture = value created * share that your firm captures

- Open standard: stronger incentives for partners, lower share captured for your firm
- Proprietary standard: weaker incentives for partners, higher share captured for your firm.

Proprietary Standards and Incentives

Do users foresee lock-in?

- This makes it hard to establish the standard.

Can you pay them off up front?...

- Penetration pricing, selective discounting, etc.

...while making sure they invest in the standard after the fact?

- ...initial payoffs are not worth it unless there is some lock-in.

Additional Issues

Competing standards means generating positive feedback quickly is essential.

- Favoring openness sooner.

Generic Strategies for standard adoption

	Control	Open
Compatible	Controlled Migration	Open Migration
Incompatible	Performance Play	Discontinuity

See Shapiro and Varian (1999).

Performance Play

Introduce high performing, incompatible product.

- Apple: Itunes, Iphone, Macintosh.

Attractive if...

- “Insanely great!” (Technology/tastes)
- Fragmented existing networks.
 - Neither you nor others have large installed bases.
- Horizontally differentiated.
- Or sufficiently ahead that can ensure migration (Nintendo?)

Controlled Migration

Compatible, but proprietary.

- IBM mainframes, Play Station, Xbox, Windows, MacOS

Attractive when...

- Compatibility valuable relative to performance.
 - Large installed base with considerable sunk-to-the-standard investments.

Open Migration

Many vendors, backwardly compatible technology.

- Televisions, fax machines, telephones, mobile phones

Attractive when...

- Compatibility valuable relative to performance.
- Contracting use of the standard easy/resolved.
 - Note these communication examples often work from de jure standards.

Discontinuity

Many vendors, new technology

- CD, DVD, HD DVD, Android.

Attractive when...

- Performance valuable relative to compatibility.
- Competing standards make quick positive feedback essential.
- Contracting w/partners over the standard relatively easy.

Discontinuity and incompatible technologies: Standards wars

Incompatible technologies battle in market in winner-take all battle

- Sony vs. Matsushita in VCR
- 3Com versus Rockwell and Lucent in 56K modems
- Digital wireless telephones: CDMA, GSM, TDMA
- Bluray, HD DVD

Strategies

Preemption

- Build an early lead, so positive feedback works for you and against rival

Expectations management

- Influence consumers purchase decisions by managing their expectations

License technology to rivals

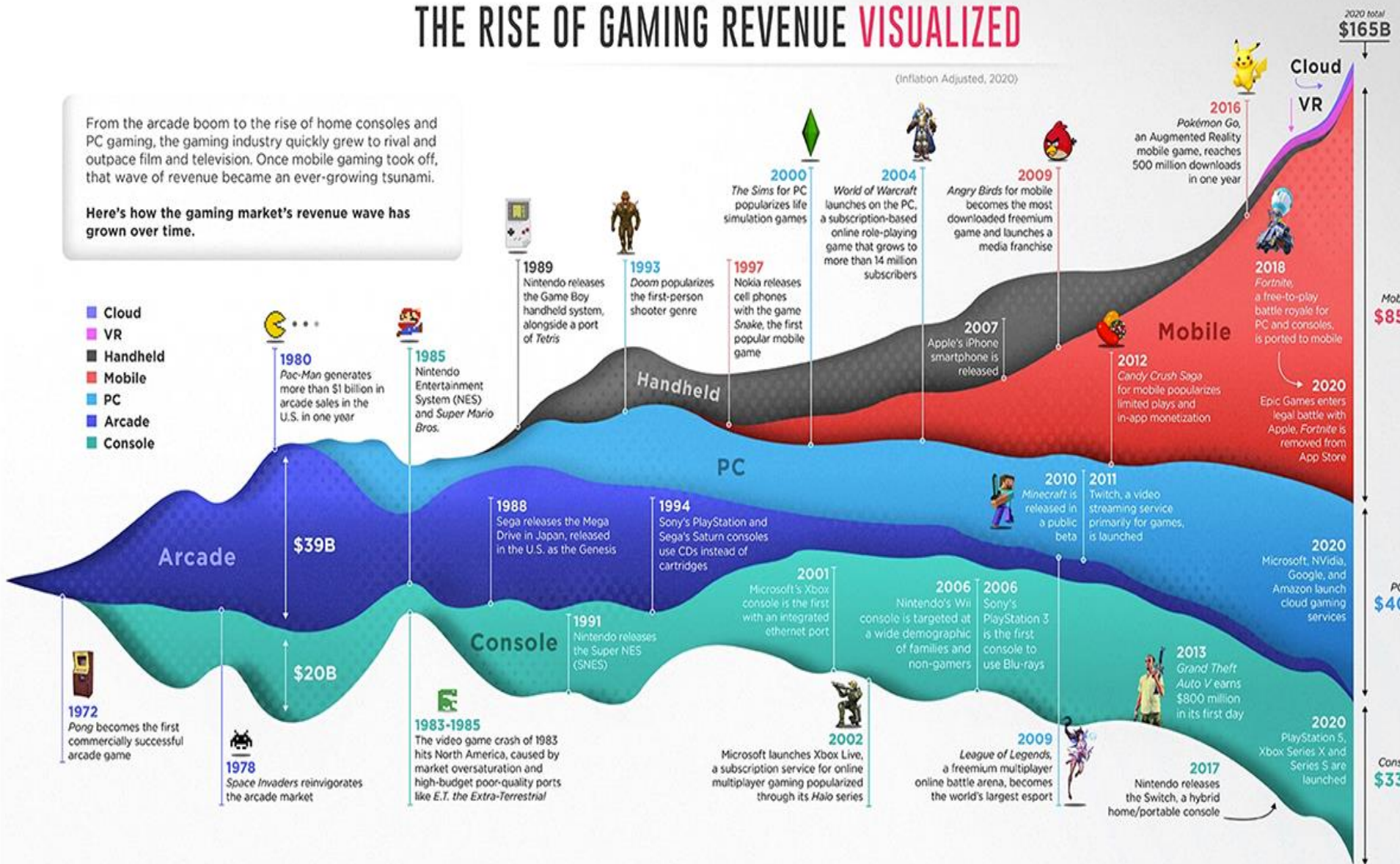
THE RISE OF GAMING REVENUE VISUALIZED

From the arcade boom to the rise of home consoles and PC gaming, the gaming industry quickly grew to rival and outpace film and television. Once mobile gaming took off, that wave of revenue became an ever-growing tsunami.

Here's how the gaming market's revenue wave has grown over time.

Legend:

- Cloud
- VR
- Handheld
- Mobile
- PC
- Arcade
- Console



Video Games

Standards issue within generation important

- games!

but compatibility across generations less important

- new games
- big technological advance
- installed base obsolete
 - large technology jumps
 - age of users

thus- standard war in each generation/ no continuity across generation

Platforms and standards

Standard is a set of technical or behavioural rules that ensure product or system compatibility.

- May emerge from market consensus (de facto) or be imposed by regulation (de jure).

Platform: an ecosystem that brings distinct user groups (for example, buyers and sellers) together and manages their interactions to generate value.

- Platforms combine a technological base with business rules and pricing strategies.

Both emerge in the presence of network externalities

Both aim to ensure compatibility

Both lead to increasing market concentration

III. Platforms

- Activities dominated by Network Effects bring people together (in a literal or figurative sense)
- Can be exchange of ideas
- Can be a transmission of data
- Can be a meeting place for buyers and sellers
- Can be physical or *digital...*
- Digitalization accounts for meteoric rise in importance
 - Internet fundamentally transforms communication
 - Cellular technology makes it omnipresent

What is new?

Platforms manage interactions and pricing across sides

Network effects, multiple equilibrium, lock in, tipping.. All still true

But now there are a few critical new aspects we shall discuss

- Platform structure (some of this holds for standard structure)
- Pricing strategies (subsidize one side to extract from the other)

Category example: Exchanges [offline]



Category example: Exchanges [online]



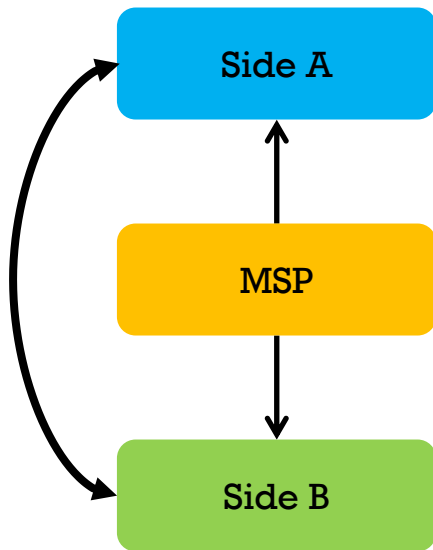
Category example: Social goods [offline]



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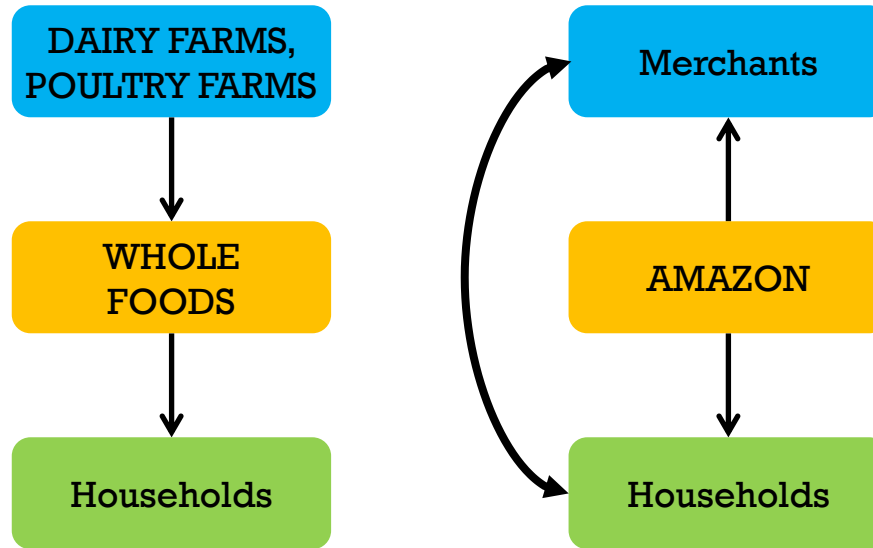


Multi-Sided Platform: Definition



- MSPs are distinguished by their **focus on the intermediary** (but intermediaries can just be resellers)
- In the platform, there is direct contact between the two sides

Definition



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- In the platform, there is direct contact between buyers and sellers

Multi-sided platforms

- The “place” where both sides “meet”
 - Generally, these are complementary goods/services
 - Often these are buyers and sellers
 - Externalities go across the platform
- Examples
 - “old economy”
 - *The New York Times*
 - *American Express*
 - “new economy”
 - *TaskRabbit*
 - *Tindr/Bumble/Grindr/Hinge/HER—take your pick*

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 - “*Old economy*”
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 - “*New economy*”
 - *TaskRabbit*
 - *Tindr/Bumble/League/Grindr/Hinge/HER—take your pick*

Managerial implications

- *Each is a clear departure from prior coursework*
(e.g., taste of Coca-Cola is not tied to how many others drink it)
- Why should managers care?
 - The problem is fundamentally different from what we've seen
 - Profit-maximizing strategies are not obvious
- The **chicken-and-egg problem** is the key challenge

How does AT&T sell the first fax machine?

How does Zuckerberg attract the first thefacebook user?

Platform Market Structure

Remember our earlier fundamentals— they are still true

1. Network Effects:

- Strong effects push toward consolidation

2. Taste for Variety:

- Strong preferences support multiple platforms

Adding platform-specific factors:

3. Scale Economies

4. Multi-homing Costs:

- Can users easily use multiple platforms?

Together, these four forces determine whether markets tip toward one platform or support multiple players.

1 Strength of Network Effects (1/2)

NE are typically not all-or-nothing feature of product or service

Are network effects important on the “extensive margin”?

- That is, does the product **function without a network**?

Are network effects important on the “intensive margin”?

- That is, does the product **function with a limited network**?
- **Strong NE on extensive margin → fewer platforms expected**
- **Strong NE on intensive margin → tipping more likely**
- [Examples follow]

1 Strength of Network Effects (2/2)

- Extensive margin
- *Can an Amex card function without any vendors?*
- *Can a Blu-Ray player function without any DVDs?*
- *Can an iPhone function without third-party apps?*

- Intensive margin
- *Can fax machines exist on a limited network?*
- *What about mobile dating apps?*
- *What about Amazon?*

2 Preferences for variety of platform function

- Relates to question we could have asked outside NE discussion: “How many product varieties do we see in equilibrium?”
- Answer depends on how heterogeneous preferences are
 - *Non-networked example: hotels vs. car wash*
 - *Networked example: dating apps vs. driving apps*
- **Strong heterog. in preferences → more platforms expected**

When talking about platforms, two more factors

3. How large are scale economies

4. How costly is to trade/interact through multiple platforms?

3 Scale economies

- Fixed costs mechanically fall per unit
 - In some industries, fixed costs dominate (e.g., perhaps total cost of producing 5MM units $\approx$ total cost of producing 10MM units)
 - Contrast manufacturing pencils vs. landline telecommunications
- *If scale econ's are large, then more/fewer platforms expected?*
- *If scale econ's are dominant, then is tipping more/less likely?*

3 Scale economies

- Fixed costs mechanically fall per unit
 - In some industries, fixed costs dominate (e.g., perhaps total cost of producing 5MM units $\approx$ total cost of producing 10MM units)
 - Contrast manufacturing pencils vs. landline telecommunications
- **Large scale economies $\rightarrow$ fewer platforms expected**
- **Scale is main cost determinant $\rightarrow$ tipping much more likely**
- Rarely do scale econ's determine platform structure
- Typically just a question of whether there is natural monopoly

4 Homing costs (1/2)

- Multi-homing costs = hassle and/or expense of maintaining presence on multiple platforms
 - Can be up-front, ongoing, at termination
 - Can be time, energy, expenditures, etc.
- Example: membership in the LinkedIn community
- **High multi-homing costs → fewer platforms expected**

[Examples follow]

4 Homing costs (2/2)

Relatively high or low multi-homing costs?

- Social media user's perspective: Facebook / Google+
- Cardholder's perspective: MasterCard / Visa / Amex.
- Household's perspective: VHS / Betamax
- User's perspective: Windows / Mac OS

Quick recap

- Platform structures
 - *Winner-take-all*
 - *Multi-homing*
 - *Mono-homing*
 - *Mixed*
- *Determinants*
 - *Strength of NE*
 - *Preference heterogeneity*
 - *Scale economies*
 - *Homing costs*

Platform structures

- **WTA** all users engage with same platform
- **Multi-homing** user interacts with >1 platform
- **Mono-homing** user interacts with 1 platform
(but diff't users choose diff't platforms)
- **Mixed homing** combination of multi-/mono-homing
- *Both sides of the market make these decisions: (e.g., you could have multi-homing on one side and mono-homing on the other)*

Platform structures

Network	Side 1 Users	Side 1 Structure	Side 2 Structure	Side 2 Users
Prof. networking sites				
Video games				
Credit cards				
Long-form video sharing				
Rural shopping malls				
Digital document formats				

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Rural shopping malls	Consumers	Mono	Multi	Stores
Digital document formats	Readers	WTA		Writers

MSP pricing

Choosing sides [1/4]

- A platform's initial concern is adoption
- Subsequent concern is equilibrium pricing
- *Notice: MSPs → multiple “sides” → multiple prices*
- Management faces a very different problem than the standard “maximum surplus extraction” problem we faced previously

Choosing sides [2/4]

- Recall from early coursework the WTP-P-C-WTS diagram
- Goal was to maximize gap between price and cost
- Wanted cheap inputs and high prices
- *Why is this much more complicated now?*

Choosing sides [3/4]

- For non-MSPs, price paid to sellers did not affect buyers and price paid to buyers did not affect sellers
- Network effects *break that irrelevance*
- *The **surplus** you extract **from one side** now affects the **surplus** you can extract **from the other side***
- With asymmetric incentives and alternatives
 - One side typically will be a loss-leader
 - One side typically will be a profit-driver

Real world MSP pricing examples

Multisided Platform	Loss leader	Profit-driver
<i>Youtube</i>		
<i>Alibaba.com, Etsy</i>		
<i>Amex, Square</i>		
<i>OpenTable</i>		
<i>PC O/S</i>		
<i>Fandango</i>		

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<i>PC O/S</i>	<i>App developers</i>	<i>Users</i>
<i>Fandango</i>	<i>Movie theatres</i>	<i>Movie goers</i>

Choosing sides [4/4]

- Setting prices when there is feedback is a complicated problem
- Viewed as an externality, it becomes more straightforward
- Example: The New York Times
 - Readers are elastic and do not care about # of advertisers
 - Advertisers are rel. inelastic and care about # of readers
- *Who would you charge high/low prices to and why?*

Choosing sides [4/4]

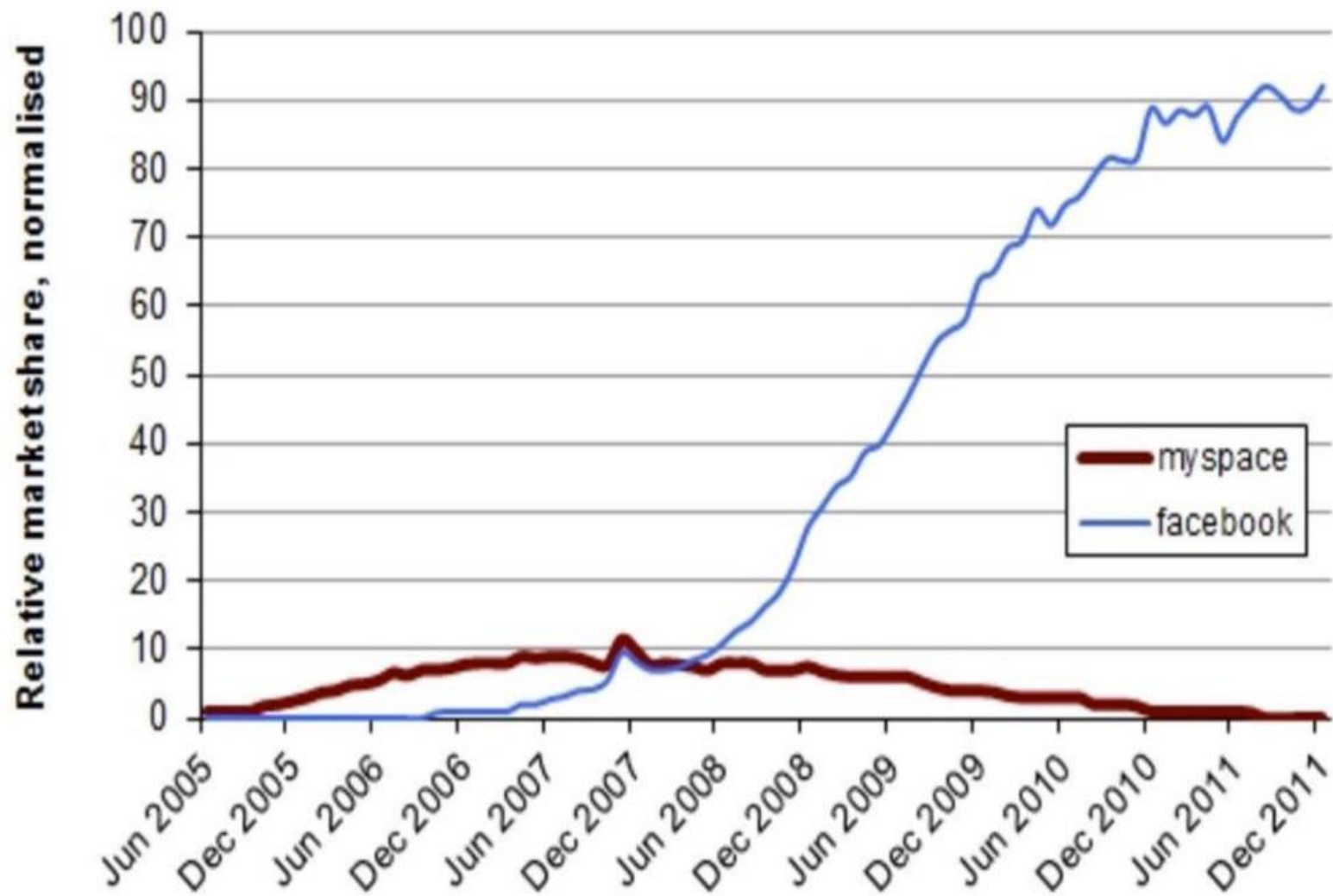
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- Example: The New York Times
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- Side that is more **elastic** = relatively **lower** prices
- Side that **generates larger indirect NE** = relatively **lower** prices
- *How “high” is the price to most credit card holders?*

Platform competition

- Forecast equilibrium platform structure to determine optimal strategic actions/reactions
- Key question: Do you expect WTA?
- Strategy framework
 - Yes → Do you have the obviously dominant technology?
 - Yes → Aggressive adoption strategy
 - No → Exit
 - Unsure → Avoid war of attrition
 - No → Turn to steady-state positioning / pricing / promotion
 - Dating apps

Market Share of Facebook vs Myspace

Global Online Social Media (2005-2011)



Wrapping Up

Network externalities are demand-side scale economies.

Results in dominant standards and platforms

A huge element of success in current economy

Compatibility is valuable, standards emerge.

- Competition between standards/ between platforms.

Multiple equilibria: inertia, positive feedback.

New trade-offs: compatibility, openness.

Coordination on adoption is life-or-death

Smart firms realize this

Pricing, entry, expectations, commitment strategies are crucial and different

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